Course Overview

Welcome to the course **Mastering Neural Networks and Model Regularization**!

This course is divided into four modules that are designed to provide practical knowledge and hands-on experience with machine learning techniques and their applications in various domains.

- Module 1: Multilayer Artificial Neural Networks
- Module 2: Model Regularization
- Module 3: Introduction to PyTorch
- Module 4: Convolutional Neural Networks

Here are the details of the modules:

- **Module 1: Multilayer Artificial Neural Networks-** In this module, you will learn about the fundamental concepts in neural networks, covering the perceptron model, model parameters, and the back-propagation algorithm. You'll also learn to implement a neural network from scratch and apply it to classify MNIST images, evaluating performance against sklearn's library function.
- **Module 2: Model Regularization-** In this module, students explore techniques to prevent overfitting in neural networks, focusing on methods such as L1 and L2 regularization and dropout. The effects of these techniques on model performance and decision boundaries will be analyzed through practical examples.
- **Module 3: Introduction to PyTorch-** This module introduces the PyTorch framework, emphasizing tensor manipulation, computational graphs, and GPU utilization. Students will gain hands-on experience in setting up and running neural networks, preparing them for advanced implementations.
- **Module 4: Convolutional Neural Networks-** Focusing on convolutional neural networks (CNNs), this module covers their architecture, operation, and applications in image and audio processing. Students will implement CNNs in PyTorch, exploring the impact of filters and pooling layers on feature extraction.

By the end of this course, students will have the skills to design, implement, and optimize multilayer perceptrons and convolutional neural networks using PyTorch, effectively apply model regularization techniques to enhance performance and analyze the results of their neural network models on various datasets.

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